Malaviya National Institute of Technology Jaipur

Department of Mathematics

B.Tech. First Semester | Odd Semester - 2025-26

End-Term Examination, November - 2025

22MAT101: MATHEMATICS - I

Duration: 2 hours 30 minutes **Max Marks:** 50

(Answer all the questions. All questions carry equal marks. Use of calculator is not allowed.)

1. Determine whether the following linear system is consistent. If it is, find a particular and a general solution; if not, justify your answer.
$$x + y - 2z + u = 1, \quad 2x + 4y - 5z + u = 5, \quad 3x + y - 5z + 5u = 3.$$

2. Consider the function $f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Show that the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ exist but the function is not continuous at the origin.

3. Using the method of Lagrange's multipliers, find the dimensions of the rectangular box of maximum volume that can be inscribed in the right circular cone $z = \sqrt{x^2 + y^2}$ with $0 \leq z \leq h$, where h is the height of the cone.

4. Show that $\displaystyle\int_0^1 \frac{x^2 dx}{\sqrt{1 - x^4}} \times \int_0^1 \frac{dx}{\sqrt{1 + x^4}} = \frac{\pi}{4\sqrt{2}}$.

5. Find the area common to the circle $r = a$ and the cardioid $r = a(1 + \cos\theta)$.

6. Evaluate the volume under the surface $f(x, y) = 8xy$ and above the region R in the first-quadrant of the xy-plane enclosed between the curves $x^2 + y^2 = 1$, $x^2 + y^2 = 4$, $x^2 - y^2 = 1$, and $x^2 - y^2 = -1$.

7. Consider the solid region V that lies in the first octant and is bounded by the surfaces $z = \sqrt{x^2 + y^2}$, $z = \sqrt{3(x^2 + y^2)}$, $z = \sqrt{1 - x^2 - y^2}$ and $z = \sqrt{4 - x^2 - y^2}$. The density of the solid at each point is inversely proportional to the square of its distance from the origin. Find the total mass of the solid.

8. Using line integral, find the lateral surface area of the portion of the right circular cylinder $x^2 + y^2 = 2$ that lies below the parabolic cylinder $z = 2 - y^2$ and above the xy-plane.

9. Verify Green's theorem for the work done by the force field $\mathbf{F}(x, y) = (x^2 - y^2)\mathbf{i} + (x + y)\mathbf{j}$ in moving a particle along the closed path C containing the curves $x + y = 0$, $x^2 + y^2 = 1$, and $x - y = 0$ in the first and fourth quadrants.

10. Verify Gauss Divergence Theorem for the vector field $\mathbf{f}(x, y, z) = xz\mathbf{i} + yz\mathbf{j} + z^2\mathbf{k}$ over the volume V bounded below by the paraboloid $z = x^2 + y^2$ and above by the portion of the upper hemisphere $z = \sqrt{2 - x^2 - y^2}$.

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Malaviya National Institute of Technology Jaipur
Department of Mathematics
22MAT101- Mathematics-I
B. Tech./Semester I/End-Term Examination 2024-25

Max. Marks: 50 Duration: 150 Minutes Date: November 29, 2024

All questions are compulsory. Calculators are NOT allowed.
Write your batch on the top right of the cover page of the answer script.

1. Find a matrix P which diagonalizes the matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Verify that $P^{-1}AP = D$, where D is a diagonal matrix. Hence find A^2.

2. Prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}$, if z is an arbitrary function of x and y and $x = e^u + e^{-v}, y = e^{-u} - e^v$.

3. What is the approximate error in the volume and surface of a rectangular parallelepiped whose length, breadth and height are measured as 10, 20 and 30 inches, respectively, if an error of -0.02 inches is made in measuring each side?

4. Find the points on the hyperbolic cylinder $x^2 - z^2 = 1$ that are closest to the origin.

5. Evaluate $\int_0^1 \frac{1}{(1-x^n)^{\frac{1}{n}}}dx$, where $n > 1$. Hence, show that $\int_0^1 \frac{1}{\sqrt[6]{1-x^6}}dx = \frac{\pi}{3}$.

6. By using polar coordinates, evaluate the double integral $\int_0^1 \int_{\frac{x}{\sqrt{3}}}^{x} \frac{x}{x^2+y^2}dydx$.

7. Find the volume of the region bounded by the cone $z = \sqrt{3(x^2+y^2)}$ and the hemisphere $z = \sqrt{4 - x^2 - y^2}$ by setting up the triple integral in spherical coordinates.

8. Prove the following:

   (a) curl $(f\vec{v})$ = (grad f) $\times \vec{v} + f$ curl $\vec{v}$

   (b) div (grad f) $= \nabla^2 f$, where $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is a Laplacian operator.

9. Verify Green's theorem for $\oint_C (xy + y^2)\, dx + x^2 dy$, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

10. Apply Gauss divergence theorem to evaluate $\iint_S \vec{F}\cdot\hat{n}\, dS$ for vector field $\vec{F} = 4x\hat{\imath} - 2y^2\hat{\jmath} + z^2\hat{k}$ taken over the region bounded by the cylinder S: $x^2 + y^2 = 4,\ z = 0,\ z = 3$.

Malaviya National Institute of Technology Jaipur
Department of Mathematics
End Term Examination: Odd Semester 2019-2020
MAT101: Mathematics-I

Remark: All questions are compulsory. Max. Mark: 50

Date: Nov. 28, 2019 Time: 2 hours 30 minutes

1. (a) Find the constants a and b such that the matrix $\begin{bmatrix} a & 4 \\ 1 & b \end{bmatrix}$ has 3 and -2 as its eigenvalues.
   (b) If $u(x,y) = \frac{1}{x^2} + \frac{1}{xy} + \frac{\log x - \log y}{x^2+y^2}$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \dots$
   (c) Express the integral $\int_0^1 x^{2/3}(1-x)^6$ in terms of beta function.
   (d) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then $\vec{E} = \vec{r}/|\vec{r}|$ is irrotational. If yes, justify your answer.
   (e) Find the constants m and n, such that the surface $3x^2 - 2y^2 - 3z^2 + 8 = 0$ is orthogonal to the surface $mx^2 + y^2 - nz = 0$ at the point $(-1, 2, 1)$. [5]

2. (a) Can there be a matrix with the eigenvectors $(1,\ 1)^T$, and $(2,\ 3)^T$ corresponding to the repeated eigenvalue 2? Justify your answer and find the matrix if possible.
   (b) State Cayley-Hamilton theorem and hence find A^n, taking $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. [2+3]

3. (a) State and prove Euler's theorem for homogeneous function of degree n in variables x, y, z.
   (b) Find the value of n, if $\theta = t^n e^{-r^2/4t}$ satisfies the relation $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$. [3+3]

4. (a) Find the length of the arc of the parabola $y^2 = 4ax$ cut off by the latus rectum.
   (b) Trace the curve $x = a\cos^3 t,\ y = a\sin^3 t$; by giving all features in detail. [2+3]

5. (a) State Gauss Divergence theorem and Stokes Theorem.
   (b) Evaluate the integral $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{1}{\sqrt{2-y^2}}\, dy dx$. [2+3]

6. If $m, n > 0$, prove that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence find the value of $\Gamma(1/2)$. [6]

7. Evaluate $\iiint_V \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2}}\, dV$; where $V = \left\{(x,y,z) : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1\right\}$. [6]

8. (a) If $\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$ and $r = |\vec{r}| \neq 0$, show that $div\,(grad\,(r^n)) = n(n+1)r^{n-2}$.
   (b) Find the directional derivative of $\nabla.\nabla\Phi$ at the point $(1,\ -2,\ 1)$ in the direction of the normal to the surface $xy^2z = 3x + z^2$, where $\Phi = 2x^3y^2z^4$. [3+3]

9. Verify Green's theorem for the vector field $\mathbf{F} = (x^3 - y^3)\hat{i} + (x+y)\hat{j}$ for the closed path C consists of the curves $x + y = 0,\ x^2 + y^2 = 1$, and $x - y = 0$ in the first and the fourth quadrant. [6]

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DEPARTMENT OF MATHEMATICS

Code :MAT101 I-Semester End term exam- 2018

Max marks: 40 Time: 2hr Date: 26/11/2018

ALL QUESTIONS CARRY EQUAL MARKS

1. Verify Gauss Divergence theorem for $\iint_S \left[\left(x^3 - yz\right)\hat{i} - 2x^2 y\hat{j} + 2\hat{k}\right] \cdot d\bar{S}$,where S denotes the surface of the cube bounded by the co-ordinate planes and the planes x=y=z=a.

2. Find the directional derivative of 1/r in the direction $\bar{r}$.

3. Derive tangential and normal acceleration of a particle describing a plane curve by Vector method.

P.T.O

$\frac{c^2}{2!}\phi_4'' + \frac{c}{2}\phi_3' + c = 0$

$\frac{c^2}{2}(0) + c(0) + c = 0$

4. Show that the larger of the two areas into which the circle $x^2 + y^2 = (8a)^2$ is divided by the parabola $y^2 = 12ax$ is $\frac{16}{3}a^2\left(8\pi - \sqrt{3}\right)$.

5. Verify the result by changing the order of integration for I= $\int_0^1 \int_{x^2}^{2-x} xy\,dx\,dy$

6. Find the characteristic equation of the matrix A. Verify Cayley-Hemilton theorem and hence compute A^{-1}, where $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

$y = x^2$

7. Verify Euler's theorem for the function $f(x,y) = \frac{x^{1/4} + y^{1/4}}{x^{1/5} + y^{1/5}}$.

8. Trace the curve $x^2 y^2 = (1+y)^2\left(4 - y^2\right)$

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Mathematics

B.Tech./I sem/2014

End-term Examination

Paper code: MAT101

Paper name: Mathematics-I

MM: 40

Time: 2 Hr

Q.1 For what values of k the equations $x+y+z=1$, $2x+y+4z=k$ and $4x+y+10z=k^2$ have a solution and solve them completely in each case. 5

Q.2 Find the constant λ so that $\mathbf{F}$ is a conservative force field where $\mathbf{F}=(\lambda xy-z^3)\hat{i}+(\lambda-2)x^2\hat{j}+(1-\lambda)xz^2\hat{k}$. Find the work done in moving particle from (1, 2, -3) to (1, -4,2). 5

Q.3 State Stoke's Theorem. Use Stoke's theorem to evaluate $\int_C (x+2y)dx+(x-z)dy+(y-z)dz$ where C is the boundary of the triangle with vertices (2,0,0), (0,3,0) and (0,0,6) oriented in the anticlockwise direction. 5

Q.4 Trace the following curve: $r=\frac{1}{2}+\cos\theta$ 5

Q.5 Find the asymptotes of the following curve: $x^3+3x^2y-4y^3-x+y+3=0$. 5

Q.6 State Euler's theorem and hence show that $x^2\frac{\partial^2 z}{\partial x^2}+2xy\frac{\partial^2 z}{\partial x\partial y}+y^2\frac{\partial^2 z}{\partial y^2}+x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=n^2 z$ where 5

$$z=x^n f_1\left(\frac{y}{x}\right)+y^{-n} f_2\left(\frac{x}{y}\right).$$

Q.7 Evaluate $\int_0^1\int_0^{\sqrt{2-y^2}}\frac{x\,dx\,dy}{\sqrt{x^2+y^2}}$ by changing the order of integration. 5

Q.8 Show that $\int_0^\infty \frac{x^2dx}{(1+x^4)^3}=\frac{5\pi\sqrt{2}}{128}$. 5